

SOLIDWORKS API (.NET VBA C#) — Reference

(317/337):

ITwoMemberCornerTreatmentFeatureData- IUserUnit

SOLIDWORKS API Documentation (converted from CHM)

ITwoMemberCornerTreatmentFeatureData (Type)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ITwoMemberCornerTreatmentFeatureData Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.sldworks Namespace](#) : ITwoMemberCornerTreatmentFeatureData Interface

Allows access to a two member corner treatment feature of a structure system.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ITwoMemberCornerTreatmentFeatureData
```

Visual Basic (Usage)



```
Dim instance As ITwoMemberCornerTreatmentFeatureData
```

C#

```
public interface ITwoMemberCornerTreatmentFeatureData
```

C++/CLI

```
public interface class ITwoMemberCornerTreatmentFeatureData
```

Visual Basic for Applications (VBA) Syntax

See [TwoMemberCornerTreatmentFeatureData](#).

Example

See the [ICornerManagementFolder](#) examples.

Remarks

This interface is:

- derived from [ICornerTreatmentFeatureData](#).
- valid only if [ICornerTreatmentFeatureData.IgnoreCornerTreatment](#) is false.

A two member corner is the position in a structure system where two members intersect. You can assign one member as a trim tool and use it to add or remove material from the other member. The trim tool member may be assigned to either member in a planar or body trim type.

In the SOLIDWORKS Design user interface, the Two member Corner PropertyManager contains three lists that you can manage using the APIs in this interface and elsewhere:

- [Corner groups](#)
- [Corner group members](#)
- [Miter trim plane point](#)

For more information, see:

- SOLIDWORKS Design user-interface help > Weldments and Structure System > Structure System > Corner Management
- SOLIDWORKS Design user-interface help > Weldments and Structure System > Structure System > Two Member PropertyManager

Accessors

Cast an ICornerTreatmentFeatureData object to this interface.

Access Diagram

[TwoMemberCornerTreatmentFeatureData](#)

See Also

[ITwoMemberCornerTreatmentFeatureData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

ITwoMemberCornerTreatmentFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)

[SolidWorks.Interop.sldworks Namespace](#) : ITwoMemberCornerTreatmentFeatureData Interface

The following tables list the members exposed by [ITwoMemberCornerTreatmentFeatureData](#).

Public Properties

Name	Description
 CornerTreatmentTrimType	Gets and sets the corner treatment trim type for this two member corner treatment feature.
 MiterTrimPlanePoint	Gets and sets the point through which to create the miter trim for this two member corner treatment feature.
 PlanarTrimOption	Gets and sets the planar trim option for this two member corner treatment feature.
 WeldTrimGap	Gets and sets the weld trim gap of this two member corner treatment feature.

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Public Methods

Name	Description
 GetCornerGroupMembers	Gets the corner group members of this two member corner treatment feature.
 SwapMembers	Assigns the trim tool functionality to the other member of this two member corner treatment feature.

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See Also

[ITwoMemberCornerTreatmentFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

ITwoMemberCornerTreatmentFeatureData Interface Methods

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : ITwoMemberCornerTreatmentFeatureData Interface

For a list of all members of this type, see [ITwoMemberCornerTreatmentFeatureData members](#).

Public Methods

Name	Description
 GetCornerGroupMembers	Gets the corner group members of this two member corner treatment feature.
 SwapMembers	Assigns the trim tool functionality to the other member of this two member corner treatment feature.

[Top](#)

See Also

[ITwoMemberCornerTreatmentFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

ITwoMemberCornerTreatmentFeatureData Interface Properties

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : ITwoMemberCornerTreatmentFeatureData Interface

For a list of all members of this type, see [ITwoMemberCornerTreatmentFeatureData members](#).

Public Properties

Name	Description
 CornerTreatmentTrimType	Gets and sets the corner treatment trim type for this two member corner treatment feature.
 MiterTrimPlanePoint	Gets and sets the point through which to create the miter trim for this two member corner treatment feature.
 PlanarTrimOption	Gets and sets the planar trim option for this two member corner treatment feature.



[WeldTrimGap](#)

Gets and sets the weld trim gap of this two member corner treatment feature.

[Top](#)

See Also

[ITwoMemberCornerTreatmentFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

CornerTreatmentTrimType Property (ITwoMemberCornerTreatmentFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ITwoMemberCornerTreatmentFeatureData Interface](#) : CornerTreatmentTrimType Property (ITwoMemberCornerTreatmentFeatureData)

Gets and sets the corner treatment trim type for this two member corner treatment feature.

.NET Syntax

Visual Basic (Declaration)

```
Property CornerTreatmentTrimType As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ITwoMemberCornerTreatmentFeatureData  
Dim value As System.Integer
```

```
instance.CornerTreatmentTrimType = value
```

```
value = instance.CornerTreatmentTrimType
```

C#

```
System.int CornerTreatmentTrimType {get; set;}
```

C++/CLI

```
property System.int CornerTreatmentTrimType {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Corner treatment trim type as defined by [swCornerTreatmentTrimType_e](#)

Visual Basic for Applications (VBA) Syntax

See [TwoMemberCornerTreatmentFeatureData::CornerTreatmentTrimType](#).

Example

See the [ICornerManagementFolder](#) examples.

See Also

[ITwoMemberCornerTreatmentFeatureData Interface](#)
[ITwoMemberCornerTreatmentFeatureData Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

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GetCornerGroupMembers Method (ITwoMemberCornerTreatmentFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ITwoMemberCornerTreatmentFeatureData Interface](#) : GetCornerGroupMembers Method (ITwoMemberCornerTreatmentFeatureData)

Gets the corner group members of this two member corner treatment feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetCornerGroupMembers() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ITwoMemberCornerTreatmentFeatureData  
Dim value As System.Object  
  
value = instance.GetCornerGroupMembers()
```

C#

```
System.Object GetCornerGroupMembers()
```

C++/CLI

```
System.Object^ GetCornerGroupMembers();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [IStructureSystemMemberFeatureData](#)s

Visual Basic for Applications (VBA) Syntax

See [TwoMemberCornerTreatmentFeatureData::GetCornerGroupMembers](#).

Example

See the [ICornerManagementFolder](#) examples.

See Also

[ITwoMemberCornerTreatmentFeatureData Interface](#)
[ITwoMemberCornerTreatmentFeatureData Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

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MiterTrimPlanePoint Property (ITwoMemberCornerTreatmentFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ITwoMemberCornerTreatmentFeatureData Interface](#) : MiterTrimPlanePoint Property (ITwoMemberCornerTreatmentFeatureData)

Gets and sets the point through which to create the miter trim for this two member corner treatment feature.

.NET Syntax

Visual Basic (Declaration)

```
Property MiterTrimPlanePoint As System.Object
```

Visual Basic (Usage)



```
Dim instance As ITwoMemberCornerTreatmentFeatureData  
Dim value As System.Object
```

```
instance.MiterTrimPlanePoint = value
value = instance.MiterTrimPlanePoint
```

C#

```
System.Object MiterTrimPlanePoint {get; set;}
```

C++/CLI

```
property System.Object^ MiterTrimPlanePoint {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[IRefPoint](#)

Visual Basic for Applications (VBA) Syntax

See [TwoMemberCornerTreatmentFeatureData::MiterTrimPlanePoint](#).

Remarks

This property is valid only if [ITwoMemberCornerTreatmentFeatureData::CornerTreatmentTrimType](#) is set to [swCornerTreatmentTrimType_e.swCornerTreatmentTrim_MiterTrim](#).

See Also

[ITwoMemberCornerTreatmentFeatureData Interface](#)
[ITwoMemberCornerTreatmentFeatureData Members](#)

Availability

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SOLIDWORKS API Help

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PlanarTrimOption Property (ITwoMemberCornerTreatmentFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ITwoMemberCornerTreatmentFeatureData Interface](#) : PlanarTrimOption Property (ITwoMemberCornerTreatmentFeatureData)

Gets and sets the planar trim option for this two member corner treatment feature.

.NET Syntax

Visual Basic (Declaration)

```
Property PlanarTrimOption As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ITwoMemberCornerTreatmentFeatureData
Dim value As System.Integer
```

```
instance.PlanarTrimOption = value
```

```
value = instance.PlanarTrimOption
```

C#

```
System.int PlanarTrimOption {get; set;}
```

C++/CLI

```
property System.int PlanarTrimOption {
    System.int get();
```

```
void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Planar trim option as defined by [swCornerTreatmentPlanarTrimOptions_e](#)

Visual Basic for Applications (VBA) Syntax

See [TwoMemberCornerTreatmentFeatureData::PlanarTrimOption](#).

Example

See the [ICornerManagementFolder](#) examples.

Remarks

This property is valid only if [ITwoMemberCornerTreatmentFeatureData::CornerTreatmentTrimType](#) is set to [swCornerTreatmentTrimType_e.swCornerTreatmentTrim_PlanarTrim](#).

See Also

[ITwoMemberCornerTreatmentFeatureData Interface](#)
[ITwoMemberCornerTreatmentFeatureData Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

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SwapMembers Method (ITwoMemberCornerTreatmentFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ITwoMemberCornerTreatmentFeatureData Interface](#) : SwapMembers Method (ITwoMemberCornerTreatmentFeatureData)

Assigns the trim tool functionality to the other member of this two member corner treatment feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub SwapMembers()
```

Visual Basic (Usage)



```
Dim instance As ITwoMemberCornerTreatmentFeatureData  
instance.SwapMembers()
```

C#

```
void SwapMembers()
```

C++/CLI

```
void SwapMembers();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [TwoMemberCornerTreatmentFeatureData::SwapMembers](#).

Example

See the [ICornerManagementFolder](#) examples.

Remarks

The first member appearing in the corner group members list in the user interface is assigned the trim tool function. Use this method to re-assign the trim tool function to the second member.

This method is valid only if [ITwoMemberCornerTreatmentFeatureData::CornerTreatmentTrimType](#) is set to [swCornerTreatmentTrimType_e](#):

- [swCornerTreatmentTrim_BodyTrim](#)
- OR -
- [swCornerTreatmentTrim_PlanarTrim](#)

See Also

[ITwoMemberCornerTreatmentFeatureData Interface](#)
[ITwoMemberCornerTreatmentFeatureData Members](#)

Availability

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SOLIDWORKS API Help

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WeldTrimGap Property (ITwoMemberCornerTreatmentFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ITwoMemberCornerTreatmentFeatureData Interface](#) : WeldTrimGap Property (ITwoMemberCornerTreatmentFeatureData)

Gets and sets the weld trim gap of this two member corner treatment feature.

.NET Syntax

Visual Basic (Declaration)

Property WeldTrimGap As System.Double

Visual Basic (Usage)



```
Dim instance As ITwoMemberCornerTreatmentFeatureData  
Dim value As System.Double
```

```
instance.WeldTrimGap = value
```

```
value = instance.WeldTrimGap
```

C#

```
System.double WeldTrimGap {get; set;}
```

C++/CLI

```
property System.double WeldTrimGap {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Weld trim gap between members

Visual Basic for Applications (VBA) Syntax

See [TwoMemberCornerTreatmentFeatureData::WeldTrimGap](#).

Example

See the [ICornerManagementFolder](#) examples.

See Also

[ITwoMemberCornerTreatmentFeatureData Interface](#)
[ITwoMemberCornerTreatmentFeatureData Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

IUniversalJointMateFeatureData (Type)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IUniversalJointMateFeatureData Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.sldworks Namespace](#) : IUniversalJointMateFeatureData Interface

Allows access to a universal joint mate feature.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IUniversalJointMateFeatureData
```

Visual Basic (Usage)



```
Dim instance As IUniversalJointMateFeatureData
```

C#

```
public interface IUniversalJointMateFeatureData
```

C++/CLI

```
public interface class IUniversalJointMateFeatureData
```

Visual Basic for Applications (VBA) Syntax

See [UniversalJointMateFeatureData](#).

Example

'VBA

' *****

' 1. Open *public_documents\samples\tutorial\api\MechanicalMates\Universal Joint.sldasm*.

' 2. Run the macro.

' 3. Inspect the Mates folder of the FeatureManager design tree.

' *****

```
Dim swApp As SldWorks.SldWorks
```

```
Dim Part As SldWorks.ModelDoc2
```

```
Dim boolstatus As Boolean
```

```
Option Explicit
```

```
Sub main()
```

```
    Set swApp = Application.SldWorks
```

```
    Set Part = swApp.ActiveDoc
```

```
    boolstatus = Part.Extension.SelectByRay(1.88633351534122E-02, 2.88569103901466E-02, 5.46569039595397E-02,  
-0.879159305813743, -0.399683188478405, -0.259484611969252, 7.73012265648448E-04, 2, True, 1, 0)
```

```
    boolstatus = Part.Extension.SelectByRay(1.86671108039604E-02, 7.47386637118552E-02, 4.19001939717987E-02,  
-0.879159305813743, -0.399683188478405, -0.259484611969252, 7.73012265648448E-04, 2, True, 1, 0)
```

```
    ' Create UniversalJointMateFeatureData
```

```
    Dim MateData As SldWorks.UniversalJointMateFeatureData
```

```
    Set MateData = Part.CreateMateData(19)
```

```
    ' Set the Entities To Mate
```

```
    Dim EntitiesToMate(1) As Object
```

```
    Set EntitiesToMate(0) = Part.SelectionManager.GetSelectedObject6(1, -1)
```

```
    Set EntitiesToMate(1) = Part.SelectionManager.GetSelectedObject6(2, -1)
```

```
    Dim EntitiesToMateVar As Variant
```

```
    EntitiesToMateVar = EntitiesToMate
```

```
    MateData.EntitiesToMate = (EntitiesToMateVar)
```

```
    ' Set the Mate DefineJointPoint
```

```
    MateData.DefineJointPoint = False
```

```
    ' Create the mate
```

```
    Dim MateFeature As SldWorks.Feature
```

```
Set MateFeature = Part.CreateMate(MateData)
Part.ClearSelection2 True
Part.EditRebuild3
End Sub
```

Remarks

A universal joint (U-joint) mate consists of two components where the rotation of one component (the output shaft) about its axis is driven by the rotation of another component (the input shaft) about its axis. An example would be the U-joint between the transmission and the rear drive axle of an automobile.

[IMateFeatureData](#) is the parent of this mate interface.

To create a universal joint mate:

- 1. Follow general instructions in the [IAssemblyDoc::CreateMate](#) Remarks.
- 2. Specify [IUniversalJointMateFeatureData::EntitiesToMate](#) or use [IModelDocExtension::SelectByID2](#) to pre-select the entities using Mark = 1.
- 3. Set [IUniversalJointMateFeatureData::DefineJointPoint](#) to true and specify [IUniversalJointMateFeatureData::JointPoint](#) or use [IModelDocExtension::SelectByID2](#) to pre-select the joint point using Mark = 16384.
- 4. Specify other properties of the UniversalJointMateFeatureData object as required.

To edit a universal joint mate:

- 1. Access its feature and call [IFeature::GetDefinition](#) to get the MateFeatureData object.
- 2. Cast the MateFeatureData object to a UniversalJointMateFeatureData object.
- 3. Modify the UniversalJointMateFeatureData object.
- 4. Call [IFeature::ModifyDefinition](#).

To delete this universal joint mate, use [IModelDocExtension::DeleteSelection2](#).

Access Diagram

[UniversalJointMateFeatureData](#)

See Also

[IUniversalJointMateFeatureData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IUniversalJointMateFeatureData Interface Members

[See Also](#) [Properties](#)



[SolidWorks.Interop.sldworks Namespace](#) : IUniversalJointMateFeatureData Interface

The following tables list the members exposed by [IUniversalJointMateFeatureData](#).

Public Properties

Name	Description
 DefineJointPoint	Gets or sets whether to define a joint point.
 EntitiesToMate	Gets or sets the entities to mate in this universal joint mate.
 JointPoint	Gets or sets the joint point of this universal joint mate.

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See Also

[IUniversalJointMateFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

IUniversalJointMateFeatureData Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IUniversalJointMateFeatureData Interface

For a list of all members of this type, see [IUniversalJointMateFeatureData members](#).

Public Properties

Name	Description
 DefineJointPoint	Gets or sets whether to define a joint point.
 EntitiesToMate	Gets or sets the entities to mate in this universal joint mate.
 JointPoint	Gets or sets the joint point of this universal joint mate.



[Top](#)

See Also

[IUniversalJointMateFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

DefineJointPoint Property (IUniversalJointMateFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUniversalJointMateFeatureData Interface](#) : DefineJointPoint Property (IUniversalJointMateFeatureData)

Gets or sets whether to define a joint point.

.NET Syntax

Visual Basic (Declaration)

Property DefineJointPoint As System.Boolean

Visual Basic (Usage)



```
Dim instance As IUniversalJointMateFeatureData  
Dim value As System.Boolean
```

```
instance.DefineJointPoint = value
```

```
value = instance.DefineJointPoint
```

C#

```
System.bool DefineJointPoint {get; set;}
```

C++/CLI

```
property System.bool DefineJointPoint {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to define a joint point, false to not

Visual Basic for Applications (VBA) Syntax

See [UniversalJointMateFeatureData::DefineJointPoint](#).

Example

See the [IUniversalJointMateFeatureData](#) example.

Remarks

If this property is set to true, then specify [IUniversalJointMateFeatureData::JointPoint](#).

See Also

[IUniversalJointMateFeatureData Interface](#)
[IUniversalJointMateFeatureData Members](#)

Availability

EntitiesToMate Property (IUniversalJointMateFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUniversalJointMateFeatureData Interface](#) : EntitiesToMate Property (IUniversalJointMateFeatureData)

Gets or sets the entities to mate in this universal joint mate.

.NET Syntax

Visual Basic (Declaration)

Property EntitiesToMate As System.Object

Visual Basic (Usage)



```
Dim instance As IUniversalJointMateFeatureData  
Dim value As System.Object
```

```
instance.EntitiesToMate = value
```

```
value = instance.EntitiesToMate
```

C#

```
System.Object EntitiesToMate {get; set;}
```

C++/CLI

```
property System.Object^ EntitiesToMate {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of two entities to mate (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [UniversalJointMateFeatureData::EntitiesToMate](#).

Example

See the [IUniversalJointMateFeatureData](#) example.

Remarks

Populate the array of this property with the cylindrical [faces](#) of two shaft components.

Instead of specifying this property, you can use [IModelDocExtension::SelectByID2](#) to pre-select the entities to mate using Mark = 1. You can pre select mate entities during mate creation, but not during mate editing.

See Also

[IUniversalJointMateFeatureData Interface](#)
[IUniversalJointMateFeatureData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

JointPoint Property (IUniversalJointMateFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUniversalJointMateFeatureData Interface](#) : JointPoint Property (IUniversalJointMateFeatureData)

Gets or sets the joint point of this universal joint mate.

.NET Syntax

Visual Basic (Declaration)

Property JointPoint As System.Object

Visual Basic (Usage)



```
Dim instance As IUniversalJointMateFeatureData
Dim value As System.Object

instance.JointPoint = value

value = instance.JointPoint
```

C#

```
System.Object JointPoint {get; set;}
```

C++/CLI

```
property System.Object^ JointPoint {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[IVertex](#) or [ISketchPoint](#) or [IRefPoint](#)

Visual Basic for Applications (VBA) Syntax

See [UniversalJointMateFeatureData::JointPoint](#).

Remarks

This property is valid only if [IUniversalJointMateFeatureData::DefineJointPoint](#) is true.

See Also

[IUniversalJointMateFeatureData Interface](#)
[IUniversalJointMateFeatureData Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

UserNotificationDefinition (Type)

IUserNotificationDefinition Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.sldworks Namespace](#) : IUserNotificationDefinition Interface

Allows access to a user notification of an add-in.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IUserNotificationDefinition
```

Visual Basic (Usage) 

```
Dim instance As IUserNotificationDefinition
```

C#

```
public interface IUserNotificationDefinition
```

C++/CLI

```
public interface class IUserNotificationDefinition
```

Visual Basic for Applications (VBA) Syntax

See [UserNotificationDefinition](#).

Example

See the [IUserNotificationHandler](#) examples.

Remarks

- A user notification is an add-in dialog that:
- Allows an add-in to display a unique message to SOLIDWORKS Design users.
 - Is modeless.
 - May appear within the context of a document open in its own frame or within SOLIDWORKS itself.
 - Replaces any previously displayed user notification.
 - Requires an IUserNotificationHandler to handle user notification events.

Accessors

[ISldWorks::DefineUserNotification](#)

Access Diagram

[UserNotificationDefinition](#)

See Also

[IUserNotificationDefinition Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMessageBarDefinition Interface](#)

IUserNotificationDefinition Interface Members

[See Also](#) [Properties](#)



[SolidWorks.Interop.sldworks Namespace](#) : IUserNotificationDefinition Interface

The following tables list the members exposed by [IUserNotificationDefinition](#).

Public Properties

Name	Description
 IncludeDoNotShowAgain	Gets or sets whether to display the "Do not show again" check box for this user notification.
 Message	Gets or sets the message displayed on this message bar.
 Position	Gets or sets the relative position of this user notification within the parent window.
 ResponseAText	Gets or sets the text displayed on the first button or command link of this user notification.
 ResponseAType	Gets or sets the first response user control for this user notification.
 ResponseBText	Gets or sets the text displayed on the second button or command link of this user notification.
 ResponseBType	Gets or sets the second response user control for this message bar.
 Severity	Gets or sets the severity level of this user notification.
 Title	Gets or sets the title of this user notification.
 UniqueName	Gets the ID of this user notification.

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See Also

[IUserNotificationDefinition Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMessageBarDefinition Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IUserNotificationDefinition Interface Properties

[See Also](#)

 [SolidWorks.Interop.sldworks Namespace](#) : IUserNotificationDefinition Interface

For a list of all members of this type, see [IUserNotificationDefinition members](#).

Public Properties

Name	Description
 IncludeDoNotShowAgain	Gets or sets whether to display the "Do not show again" check box for this user notification.
 Message	Gets or sets the message displayed on this message bar.
 Position	Gets or sets the relative position of this user notification within the parent window.
 ResponseAText	Gets or sets the text displayed on the first button or command link of this user notification.
 ResponseAType	Gets or sets the first response user control for this user notification.
 ResponseBText	Gets or sets the text displayed on the second button or command link of this user notification.
 ResponseBType	Gets or sets the second response user control for this message bar.
 Severity	Gets or sets the severity level of this user notification.
 Title	Gets or sets the title of this user notification.
 UniqueName	Gets the ID of this user notification.

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See Also

[IUserNotificationDefinition Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMessageBarDefinition Interface](#)

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IncludeDoNotShowAgain Property (IUserNotificationDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserNotificationDefinition Interface](#) : IncludeDoNotShowAgain Property (IUserNotificationDefinition)

Gets or sets whether to display the "Do not show again" check box for this user notification.

.NET Syntax

Visual Basic (Declaration)

Property IncludeDoNotShowAgain As System.Boolean

Visual Basic (Usage)



```
Dim instance As IUserNotificationDefinition
Dim value As System.Boolean

instance.IncludeDoNotShowAgain = value
value = instance.IncludeDoNotShowAgain
```

C#

```
System.bool IncludeDoNotShowAgain {get; set;}
```

C++/CLI

```
property System.bool IncludeDoNotShowAgain {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True (default) to display the check box, false to not.

Visual Basic for Applications (VBA) Syntax

See [UserNotificationDefinition::IncludeDoNotShowAgain](#).

See Also

[IUserNotificationDefinition Interface](#)
[IUserNotificationDefinition Members](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments](#) on this topic.

Message Property (IUserNotificationDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserNotificationDefinition Interface](#) : Message Property (IUserNotificationDefinition)

Gets or sets the message displayed on this message bar.

.NET Syntax

Visual Basic (Declaration)

Property Message As System.String

Visual Basic (Usage)



```
Dim instance As IUserNotificationDefinition
Dim value As System.String

instance.Message = value
value = instance.Message
```

C#

```
System.string Message {get; set;}
```

C++/CLI

```
property System.String^ Message {  
    System.String^ get();  
    void set ( & System.String^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Non-empty message string

Visual Basic for Applications (VBA) Syntax

See [UserNotificationDefinition::Message](#).

Example

See the [IUserNotificationHandler](#) examples.

Remarks

This property must be set.

See Also

[IUserNotificationDefinition Interface](#)
[IUserNotificationDefinition Members](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

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Position Property (IUserNotificationDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserNotificationDefinition Interface](#) : Position Property (IUserNotificationDefinition)

Gets or sets the relative position of this user notification within the parent window.

.NET Syntax

Visual Basic (Declaration)

```
Property Position As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IUserNotificationDefinition  
Dim value As System.Integer
```

```
instance.Position = value
```

```
value = instance.Position
```

C#

```
System.int Position {get; set;}
```

C++/CLI

```
property System.int Position {  
    System.int get();
```

```
void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Position as defined by [swUserNotificationPosition_e](#)

Visual Basic for Applications (VBA) Syntax

See [UserNotificationDefinition::Position](#).

Remarks

Default value of this property is swUserNotificationPosition_e.swUserNotificationPosition_Default.

See Also

[IUserNotificationDefinition Interface](#)
[IUserNotificationDefinition Members](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

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ResponseAText Property (IUserNotificationDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserNotificationDefinition Interface](#) : ResponseAText Property (IUserNotificationDefinition)

Gets or sets the text displayed on the first button or command link of this user notification.

.NET Syntax

Visual Basic (Declaration)

```
Property ResponseAText As System.String
```

Visual Basic (Usage)



```
Dim instance As IUserNotificationDefinition  
Dim value As System.String  
  
instance.ResponseAText = value  
value = instance.ResponseAText
```

C#

```
System.string ResponseAText {get; set;}
```

C++/CLI

```
property System.String^ ResponseAText {  
    System.String^ get();  
    void set ( & System.String^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Text displayed on first user response control; default is an empty string

Visual Basic for Applications (VBA) Syntax

See [UserNotificationDefinition::ResponseAText](#).

Example

See the [IUserNotificationHandler](#) examples.

See Also

[IUserNotificationDefinition Interface](#)
[IUserNotificationDefinition Members](#)

Availability

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ResponseAType Property (IUserNotificationDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserNotificationDefinition Interface](#) : ResponseAType Property (IUserNotificationDefinition)

Gets or sets the first response user control for this user notification.

.NET Syntax

Visual Basic (Declaration)

Property ResponseAType As System.Integer

Visual Basic (Usage)



```
Dim instance As IUserNotificationDefinition  
Dim value As System.Integer
```

```
instance.ResponseAType = value
```

```
value = instance.ResponseAType
```

C#

```
System.int ResponseAType {get; set;}
```

C++/CLI

```
property System.int ResponseAType {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

First response user control as defined by [swUserNotificationResponseType_e](#)

Visual Basic for Applications (VBA) Syntax

See [UserNotificationDefinition::ResponseAType](#).

Example

See the [IUserNotificationHandler](#) examples.

See Also

[IUserNotificationDefinition Interface](#)
[IUserNotificationDefinition Members](#)

Availability

ResponseBText Property (IUserNotificationDefinition)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IUserNotificationDefinition Interface](#) : ResponseBText Property (IUserNotificationDefinition)

Gets or sets the text displayed on the second button or command link of this user notification.

.NET Syntax

Visual Basic (Declaration)

Property ResponseBText As System.String

Visual Basic (Usage)



```
Dim instance As IUserNotificationDefinition
Dim value As System.String
```

```
instance.ResponseBText = value
```

```
value = instance.ResponseBText
```

C#

```
System.string ResponseBText {get; set;}
```

C++/CLI

```
property System.String^ ResponseBText {
    System.String^ get();
    void set ( & System.String^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Text displayed on second user response control; default is an empty string

Visual Basic for Applications (VBA) Syntax

See [UserNotificationDefinition::ResponseBText](#).

See Also

[IUserNotificationDefinition Interface](#)
[IUserNotificationDefinition Members](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

ResponseBType Property (IUserNotificationDefinition)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IUserNotificationDefinition Interface](#) : ResponseBType Property (IUserNotificationDefinition)

Gets or sets the second response user control for this message bar.

.NET Syntax

Visual Basic (Declaration)

Property ResponseBType As System.Integer

Visual Basic (Usage)



```
Dim instance As IUserNotificationDefinition
Dim value As System.Integer

instance.ResponseBType = value
value = instance.ResponseBType
```

C#

```
System.int ResponseBType {get; set;}
```

C++/CLI

```
property System.int ResponseBType {
    System.int get();
    void set ( &    System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Second response user control as defined by [swUserNotificationResponseType_e](#)

Visual Basic for Applications (VBA) Syntax

See [UserNotificationDefinition::ResponseBType](#).

See Also

[IUserNotificationDefinition Interface](#)
[IUserNotificationDefinition Members](#)

Availability

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SOLIDWORKS API Help

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Severity Property (IUserNotificationDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserNotificationDefinition Interface](#) : Severity Property (IUserNotificationDefinition)

Gets or sets the severity level of this user notification.

.NET Syntax

Visual Basic (Declaration)

```
Property Severity As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IUserNotificationDefinition
Dim value As System.Integer

instance.Severity = value
value = instance.Severity
```

C#

```
System.int Severity {get; set;}
```

C++/CLI

```
property System.int Severity {
    System.int get();
```

```
void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Severity level as defined by [swUserNotificationSeverity_e](#)

Visual Basic for Applications (VBA) Syntax

See [UserNotificationDefinition::Severity](#).

Example

See the [IUserNotificationHandler](#) examples.

See Also

[IUserNotificationDefinition Interface](#)
[IUserNotificationDefinition Members](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

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Title Property (IUserNotificationDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserNotificationDefinition Interface](#) : Title Property (IUserNotificationDefinition)

Gets or sets the title of this user notification.

.NET Syntax

Visual Basic (Declaration)

```
Property Title As System.String
```

Visual Basic (Usage)



```
Dim instance As IUserNotificationDefinition  
Dim value As System.String  
  
instance.Title = value  
  
value = instance.Title
```

C#

```
System.string Title {get; set;}
```

C++/CLI

```
property System.String^ Title {  
    System.String^ get();  
    void set ( & System.String^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Non-empty title string

Visual Basic for Applications (VBA) Syntax

See [UserNotificationDefinition::Title](#).

Example

See the [IUserNotificationHandler](#) examples.

Remarks

This property must be set.

See Also

[IUserNotificationDefinition Interface](#)
[IUserNotificationDefinition Members](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

UniqueName Property (IUserNotificationDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserNotificationDefinition Interface](#) : UniqueName Property (IUserNotificationDefinition)

Gets the ID of this user notification.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property UniqueName As System.String
```

Visual Basic (Usage)



```
Dim instance As IUserNotificationDefinition  
Dim value As System.String  
  
value = instance.UniqueName
```

C#

```
System.string UniqueName {get;}
```

C++/CLI

```
property System.String^ UniqueName {  
    System.String^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Unique ID of this user notification

Visual Basic for Applications (VBA) Syntax

See [UserNotificationDefinition::UniqueName](#).

Remarks

It is the add-in's responsibility to ensure that the ID is unique by using, for example, a combination of add-in module name and unique notification identifier:

"MyAddInName+ID_MYNOTIFICATION1"

See Also

[IUserNotificationDefinition Interface](#)
[IUserNotificationDefinition Members](#)

IUserProgressBar (Type)

IUserProgressBar Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IUserProgressBar Interface

Allows access to a progress indicator that indicates how much longer a SOLIDWORKS operation will take.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IUserProgressBar
```

Visual Basic (Usage)



```
Dim instance As IUserProgressBar
```

C#

```
public interface IUserProgressBar
```

C++/CLI

```
public interface class IUserProgressBar
```

Visual Basic for Applications (VBA) Syntax

See [UserProgressBar](#).

Example

[Start, Update, and Stop Progress Indicator \(VBA\)](#)
[Start, Update, and Stop Progress Indicator \(VB.NET\)](#)
[Start, Update, and Stop Progress Indicator \(C#\)](#)

Remarks

The progress indicator appears on the SOLIDWORKS status bar.

Accessors

[ISldWorks::GetUserProgressBar](#)

Access Diagram

[UserProgressBar](#)

See Also

[IUserProgressBar Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IUserProgressBar Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IUserProgressBar Interface

For a list of all members of this type, see [IUserProgressBar members](#).

Public Methods

Name	Description
 End	Removes the progress indicator from the SOLIDWORKS status bar.
 Start	Sets the range of the progress indicator display and shows it on the SOLIDWORKS status bar.
 UpdateProgress	Increments the progress indicator.
 UpdateTitle	Changes the title of the progress indicator.

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 See Also

[IUserProgressBar Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IUserProgressBar Interface Members

[See Also](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IUserProgressBar Interface

The following tables list the members exposed by [IUserProgressBar](#).

 Public Methods

Name	Description
 End	Removes the progress indicator from the SOLIDWORKS status bar.
 Start	Sets the range of the progress indicator display and shows it on the SOLIDWORKS status bar.
 UpdateProgress	Increments the progress indicator.
 UpdateTitle	Changes the title of the progress indicator.

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 See Also

[IUserProgressBar Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

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End Method (IUserProgressBar)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserProgressBar Interface](#) : End Method (IUserProgressBar)

Removes the progress indicator from the SOLIDWORKS status bar.

 C# Syntax

Visual Basic (Declaration)

```
Function End() As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IUserProgressBar  
Dim value As System.Boolean  
  
value = instance.End()
```

C#

```
System.bool End()
```

C++/CLI

```
System.bool End();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the progress indicator is removed, false if not

Visual Basic for Applications (VBA) Syntax

See [UserProgressBar::End](#).

Example

See the [IUserProgressBar](#) examples.

Remarks

This method does not immediately remove the progress bar from the status bar. Instead, a show-window event is posted to the message queue of the given window when this method is called. After Windows processes the message, the progress bar is removed from the status bar.

See Also

[IUserProgressBar Interface](#)
[IUserProgressBar Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Start Method (IUserProgressBar)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserProgressBar Interface](#) : Start Method (IUserProgressBar)

Sets the range of the progress indicator display and shows it on the SOLIDWORKS status bar.

.NET Syntax

Visual Basic (Declaration)

```
Function Start( _  
    ByVal LowerBound As System.Integer, _  
    ByVal UpperBound As System.Integer, _  
    ByVal ProgressBarTitle As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IUserProgressBar  
Dim LowerBound As System.Integer  
Dim UpperBound As System.Integer  
Dim ProgressBarTitle As System.String  
Dim value As System.Boolean  
  
value = instance.Start(LowerBound, UpperBound, ProgressBarTitle)
```

C#

```
System.bool Start(  
    System.int LowerBound,  
    System.int UpperBound,  
    System.string ProgressBarTitle  
)
```

C++/CLI

```
System.bool Start(  
& System.int LowerBound,  
& System.int UpperBound,  
& System.String^ ProgressBarTitle  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

LowerBound
Lower bound of range
UpperBound
Upper bound of range
ProgressBarTitle
Title of progress indicator to show in status bar

Return Value

True if progress indicator successfully started, false if not

Visual Basic for Applications (VBA) Syntax

See [UserProgressBar::Start](#).

Example

See the [IUserProgressBar](#) examples.

See Also

[IUserProgressBar Interface](#)
[IUserProgressBar Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

UpdateProgress Method (IUserProgressBar)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserProgressBar Interface](#) : UpdateProgress Method (IUserProgressBar)

Increments the progress indicator.

.NET Syntax

Visual Basic (Declaration)

```
Function UpdateProgress( _  
    ByVal Position As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IUserProgressBar  
Dim Position As System.Integer  
Dim value As System.Integer  
  
value = instance.UpdateProgress(Position)
```

C#

```
System.int UpdateProgress(  
    System.int Position  
)
```

C++/CLI



```
System.int UpdateProgress(  
& System.int Position  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Position
New position of the progress indicator

Return Value

Status of progress indicator update as defined in [swUpdateProgressError_e](#)

Visual Basic for Applications (VBA) Syntax

See [UserProgressBar::UpdateProgress](#).

Example

See the [IUserProgressBar](#) examples.

See Also

[IUserProgressBar Interface](#)
[IUserProgressBar Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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UpdateTitle Method (IUserProgressBar)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserProgressBar Interface](#) : UpdateTitle Method (IUserProgressBar)

Changes the title of the progress indicator.

.NET Syntax

Visual Basic (Declaration)

```
Function UpdateTitle( _  
    ByVal ProgressBarTitle As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IUserProgressBar  
Dim ProgressBarTitle As System.String  
Dim value As System.Boolean  
  
value = instance.UpdateTitle(ProgressBarTitle)
```

C#

```
System.bool UpdateTitle(  
    System.string ProgressBarTitle  
)
```

C++/CLI

```
System.bool UpdateTitle(  
& System.String^ ProgressBarTitle  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ProgressBarTitle
New title of progress indicator

Return Value

True if title successfully updated, false if not

Visual Basic for Applications (VBA) Syntax

See [UserProgressBar::UpdateTitle](#).

Example

See the [UserProgressBar](#) examples.

See Also

[IUserProgressBar Interface](#)
[IUserProgressBar Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

IUserUnit (Type)

IUserUnit Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IUserUnit Interface

Allows you to manage units.
NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IUserUnit
```

Visual Basic (Usage)



```
Dim instance As IUserUnit
```

C#

```
public interface IUserUnit
```

C++/CLI

```
public interface class IUserUnit
```

Visual Basic for Applications (VBA) Syntax

See [UserUnit](#).

Example

[Get and Set Document Units \(VBA\)](#)
[Get and Set Document Units \(VB.NET\)](#)
[Get and Set Document Units \(C#\)](#)

Remarks

If you obtain IUserUnit using...	Then IUserUnit's properties are...
IModelDoc2	Read only and persistent. Use IModelDocExtension::SetUserPreference* methods to set the units properties of a document.
ISldWorks	Read-write, but not persistent. The instance of IUserUnit returned by ISldWorks is empty and not tied to any document. Use this instance as a template to store units properties at runtime.

NOTE: Use this interface instead of **mo_UserUnits.h** file in *public_documents\lappcomm* of your SOLIDWORKS installation. If you previously used **mo_UserUnits.h**, you should change your applications to use this interface.

Accessors

[IModelDoc2::GetUserUnit](#) and [IModelDoc2::!GetUserUnit](#)
[ISldWorks::GetUserUnit](#) and [ISldWorks::!GetUserUnit](#)

Access Diagram

[UserUnit](#)

See Also

[IUserUnit Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IUserUnit Interface Methods

[See Also](#)



For a list of all members of this type, see [IUserUnit members](#).

Public Methods

Name	Description
 ConvertDoubleToSystemValue	Converts a double value to a document unit value.
 ConvertToSystemValue	Converts a text string to a document unit value.
 ConvertToUserUnit	Converts the input value to document units.
 GetConversionFactor	Gets the conversion factor.
 GetFullUnitName	Gets the full name of the unit.
 GetUnitsString	Gets the string that describes the unit.
 GetUserAngleTolerance	Gets the value of the angle tolerance for an angle unit.
 IsMetric	Gets whether the unit is metric.

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See Also

[IUserUnit Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

IUserUnit Interface Members

[See Also](#) [Properties](#) [Methods](#)



The following tables list the members exposed by [IUserUnit](#).

Public Properties

Name	Description
 DisplayLeadingZero	Gets whether to display leading zeroes.
 FractionBase	Gets the fraction base.
 FractionValue	Gets the denominator value of the fraction.
 PadZero	Gets whether to pad with zeroes.
 RoundToFraction	Gets whether to round to a fraction.
 SeparatorCharacter	Gets the decimal separator character.
 SignificantDigits	Gets the number of significant digits.
 SpecificUnitType	Gets the specific unit type for this user unit.
 UnitType	Gets the user unit type.

[Top](#)

Public Methods

Name	Description
 ConvertDoubleToSystemValue	Converts a double value to a document unit value.
 ConvertToSystemValue	Converts a text string to a document unit value.
 ConvertToUserUnit	Converts the input value to document units.

 GetConversionFactor	Gets the conversion factor.
 GetFullUnitName	Gets the full name of the unit.
 GetUnitsString	Gets the string that describes the unit.
 GetUserAngleTolerance	Gets the value of the angle tolerance for an angle unit.
 IsMetric	Gets whether the unit is metric.

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 See Also

[IUserUnit Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help [Send comments on this topic.](#)

IUserUnit Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IUserUnit Interface

For a list of all members of this type, see [IUserUnit members](#).

 Public Properties

Name	Description
 DisplayLeadingZero	Gets whether to display leading zeroes.
 FractionBase	Gets the fraction base.
 FractionValue	Gets the denominator value of the fraction.
 PadZero	Gets whether to pad with zeroes.
 RoundToFraction	Gets whether to round to a fraction.
 SeparatorCharacter	Gets the decimal separator character.
 SignificantDigits	Gets the number of significant digits.
 SpecificUnitType	Gets the specific unit type for this user unit.
 UnitType	Gets the user unit type.

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 See Also

[IUserUnit Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help [Send comments on this topic.](#)

ConvertDoubleToSystemValue Method (IUserUnit)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserUnit Interface](#) : ConvertDoubleToSystemValue Method (IUserUnit)

Converts a double value to a document unit value.

 .NET Syntax

Visual Basic (Declaration)

```
Function ConvertDoubleToSystemValue( _  
    ByVal UserValue As System.Double _  
) As System.Double
```

```
Dim instance As IUserUnit
Dim UserValue As System.Double
Dim value As System.Double

value = instance.ConvertDoubleToSystemValue(UserValue)
```

C#

```
System.double ConvertDoubleToSystemValue(
    System.double UserValue
)
```

C++/CLI

```
System.double ConvertDoubleToSystemValue(
    & System.double UserValue
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters*UserValue*

Value to convert

Return Value

Converted document unit value

Visual Basic for Applications (VBA) SyntaxSee [UserUnit::ConvertDoubleToSystemValue](#). **Example**See the [IUserUnit](#) examples. **See Also**

[IUserUnit Interface](#)
[IUserUnit Members](#)
[IUserUnit::ConvertToSystemValue Method](#)
[IUserUnit::ConvertToUserUnit Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments](#) on this topic.**ConvertToSystemValue Method (IUserUnit)**[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IUserUnit Interface](#) : ConvertToSystemValue Method (IUserUnit)

Converts a text string to a document unit value.

.NET Syntax

Visual Basic (Declaration)

```
Function ConvertToSystemValue( _
    ByVal UnitText As System.String, _
    ByRef ComputedValue As System.Double _
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IUserUnit
Dim UnitText As System.String
Dim ComputedValue As System.Double
Dim value As System.Boolean

value = instance.ConvertToSystemValue(UnitText, ComputedValue)
```

C#

```
System.bool ConvertToSystemValue(
    System.string UnitText,
    out System.double ComputedValue
)
```

C++/CLI

```
System.bool ConvertToSystemValue(
    & System.String^ UnitText,
    & [Out] System.double ComputedValue
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UnitText

Value to convert

ComputedValue

Converted value in document units

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [UserUnit::ConvertToSystemValue](#).

Example

See the [IUserUnit](#) examples.

Remarks

This takes the value directly from a textbox control and passes it in to convert to a double value.

This imitates the behavior of when working in the SOLIDWORKS Design user interface and passing a value like "10 / 2" into one of the values of a dialog box.

See Also

[IUserUnit Interface](#)

[IUserUnit Members](#)

[IUserUnit::ConvertDoubleToSystemValue Method](#)

[IUserUnit::ConvertToUserUnit Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ConvertToUserUnit Method (IUserUnit)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserUnit Interface](#) : ConvertToUserUnit Method (IUserUnit)

Converts the input value to document units.

.NET Syntax

Visual Basic (Declaration)

```
Function ConvertToUserUnit( _  
    ByVal ValueIn As System.Double, _  
    ByVal ShowUsernames As System.Boolean, _  
    ByVal NameInEnglish As System.Boolean _  
    ) As System.String
```

Visual Basic (Usage)



```
Dim instance As IUserUnit  
Dim ValueIn As System.Double  
Dim ShowUsernames As System.Boolean  
Dim NameInEnglish As System.Boolean  
Dim value As System.String  
  
value = instance.ConvertToUserUnit(ValueIn, ShowUsernames, NameInEnglish)
```

C#

```
System.string ConvertToUserUnit(  
    System.double ValueIn,  
    System.bool ShowUsernames,  
    System.bool NameInEnglish  
)
```

C++/CLI

```
System.String^ ConvertToUserUnit(  
& System.double ValueIn,  
& System.bool ShowUsernames,  
& System.bool NameInEnglish  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ValueIn

Value to convert

ShowUsernames

True to show the user names, false to not

NameInEnglish

True to return the name in English, false to not

Return Value

Conversion to a string

Visual Basic for Applications (VBA) Syntax

See [UserUnit::ConvertToUserUnit](#).

Example

See the [IUserUnit](#) examples.

See Also

[IUserUnit Interface](#)

[IUserUnit Members](#)

[IUserUnit::ConvertDoubleToSystemValue Method](#)

[IUserUnit::ConvertToSystemValue Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserUnit Interface](#) : DisplayLeadingZero Property (IUserUnit)

Gets whether to display leading zeroes.

.NET Syntax

Visual Basic (Declaration)

Property DisplayLeadingZero As System.Boolean

Visual Basic (Usage)



```
Dim instance As IUserUnit
Dim value As System.Boolean

instance.DisplayLeadingZero = value

value = instance.DisplayLeadingZero
```

C#

```
System.bool DisplayLeadingZero {get; set;}
```

C++/CLI

```
property System.bool DisplayLeadingZero {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to display the leading zero, false to not

Visual Basic for Applications (VBA) Syntax

See [UserUnit.DisplayLeadingZero](#).

Example

See the [IUserUnit](#) examples.

See Also

[IUserUnit Interface](#)
[IUserUnit Members](#)

Availability

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SOLIDWORKS API Help

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FractionBase Property (IUserUnit)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserUnit Interface](#) : FractionBase Property (IUserUnit)

Gets the fraction base.

.NET Syntax

Visual Basic (Declaration)

Property FractionBase As System.Integer

Visual Basic (Usage)

```
Dim instance As IUserUnit
Dim value As System.Integer

instance.FractionBase = value
value = instance.FractionBase
```

C#

```
System.int FractionBase {get; set;}
```

C++/CLI

```
property System.int FractionBase {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Base as defined in [swFractionDisplay_e](#)

Visual Basic for Applications (VBA) Syntax

See [UserUnit::FractionBase](#).

Example

See the [IUserUnit](#) examples.

See Also

[IUserUnit Interface](#)
[IUserUnit Members](#)

Availability

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FractionValue Property (IUserUnit)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserUnit Interface](#) : FractionValue Property (IUserUnit)

Gets the denominator value of the fraction.

.NET Syntax

Visual Basic (Declaration)

```
Property FractionValue As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IUserUnit
Dim value As System.Integer

instance.FractionValue = value
value = instance.FractionValue
```

C#

```
System.int FractionValue {get; set;}
```

C++/CLI

```
property System.int FractionValue {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Fraction denominator value; valid only if [IUserUnit::FractionBase](#) is [swFractionDisplay_e.swFRACTION](#)

Visual Basic for Applications (VBA) Syntax

See [UserUnit::FractionValue](#).

Example

See the [IUserUnit](#) examples.

See Also

[IUserUnit Interface](#)
[IUserUnit Members](#)

Availability

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SOLIDWORKS API Help

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GetConversionFactor Method (IUserUnit)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserUnit Interface](#) : GetConversionFactor Method (IUserUnit)

Gets the conversion factor.

.NET Syntax

Visual Basic (Declaration)

```
Function GetConversionFactor() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IUserUnit  
Dim value As System.Double  
  
value = instance.GetConversionFactor()
```

C#

```
System.double GetConversionFactor()
```

C++/CLI

```
System.double GetConversionFactor();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Conversion factor

Visual Basic for Applications (VBA) Syntax

See [UserUnit::GetConversionFactor](#).

Example

See the [IUserUnit](#) examples.

See Also

[IUserUnit Interface](#)
[IUserUnit Members](#)

Availability

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SOLIDWORKS API Help

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GetFullUnitName Method (IUserUnit)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserUnit Interface](#) : GetFullUnitName Method (IUserUnit)

Gets the full name of the unit.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFullUnitName( _  
    ByVal Plural As System.Boolean _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IUserUnit  
Dim Plural As System.Boolean  
Dim value As System.String  
  
value = instance.GetFullUnitName(Plural)
```

C#

```
System.string GetFullUnitName(  
    System.bool Plural  
)
```

C++/CLI

```
System.String^ GetFullUnitName(  
&    System.bool Plural  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Plural

True if the name of the unit is plural, false if not

Return Value

Full name of the unit

Visual Basic for Applications (VBA) Syntax

See [UserUnit::GetFullUnitName](#).

Example

See the [IUserUnit](#) examples.

See Also

[IUserUnit Interface](#)
[IUserUnit Members](#)

Availability

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GetUnitsString Method (IUserUnit)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserUnit Interface](#) : GetUnitsString Method (IUserUnit)

Gets the string that describes the unit.

.NET Syntax

Visual Basic (Declaration)

```
Function GetUnitsString( _  
    ByVal InEnglish As System.Boolean _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IUserUnit  
Dim InEnglish As System.Boolean  
Dim value As System.String  
  
value = instance.GetUnitsString(InEnglish)
```

C#

```
System.string GetUnitsString(  
    System.bool InEnglish  
)
```

C++/CLI

```
System.String^ GetUnitsString(  
& System.bool InEnglish  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

InEnglish

True to return the string in English, false to not

Return Value

String describing the unit

Visual Basic for Applications (VBA) Syntax

See [UserUnit::GetUnitsString](#).

Example

See the [IUserUnit](#) examples.

See Also

[IUserUnit Interface](#)
[IUserUnit Members](#)

Availability

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SOLIDWORKS API Help

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GetUserAngleTolerance Method (IUserUnit)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserUnit Interface](#) : GetUserAngleTolerance Method (IUserUnit)

Gets the value of the angle tolerance for an angle unit.

.NET Syntax

Visual Basic (Declaration)

```
Function GetUserAngleTolerance() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IUserUnit  
Dim value As System.Double
```

```
value = instance.GetUserAngleTolerance()
```

C#

```
System.double GetUserAngleTolerance()
```

C++/CLI

```
System.double GetUserAngleTolerance();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Value of the angle tolerance

Visual Basic for Applications (VBA) Syntax

See [UserUnit.GetUserAngleTolerance](#).

Example

See the [IUserUnit](#) examples.

See Also

[IUserUnit Interface](#)
[IUserUnit Members](#)

Availability

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SOLIDWORKS API Help

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IsMetric Method (IUserUnit)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserUnit Interface](#) : IsMetric Method (IUserUnit)

Gets whether the unit is metric.

.NET Syntax

Visual Basic (Declaration)

Function IsMetric() As System.Boolean

Visual Basic (Usage)



```
Dim instance As IUserUnit
Dim value As System.Boolean

value = instance.IsMetric()
```

C#

```
System.bool IsMetric()
```

C++/CLI

```
System.bool IsMetric();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the unit is metric, false if not

Visual Basic for Applications (VBA) Syntax

See [UserUnit::IsMetric](#).

Example

See the [IUserUnit](#) examples.

See Also

[IUserUnit Interface](#)
[IUserUnit Members](#)

Availability

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PadZero Property (IUserUnit)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserUnit Interface](#) : PadZero Property (IUserUnit)

Gets whether to pad with zeroes.

.NET Syntax

Visual Basic (Declaration)

```
Property PadZero As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IUserUnit
Dim value As System.Boolean

instance.PadZero = value

value = instance.PadZero
```

C#

```
System.bool PadZero {get; set;}
```

C++/CLI

```
property System.bool PadZero {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to pad with zeroes, false to not

Visual Basic for Applications (VBA) Syntax

See [UserUnit.PadZero](#).

Example

See the [IUserUnit](#) examples.

See Also

[IUserUnit Interface](#)
[IUserUnit Members](#)

Availability

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RoundToFraction Property (IUserUnit)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserUnit Interface](#) : RoundToFraction Property (IUserUnit)

Gets whether to round to a fraction.

.NET Syntax

Visual Basic (Declaration)

```
Property RoundToFraction As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IUserUnit
Dim value As System.Boolean

instance.RoundToFraction = value
value = instance.RoundToFraction
```

C#

```
System.bool RoundToFraction {get; set;}
```

C++/CLI

```
property System.bool RoundToFraction {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to round, false to not

Visual Basic for Applications (VBA) Syntax

See [UserUnit.RoundToFraction](#).

Example

See the [IUserUnit](#) examples.

See Also

[IUserUnit Interface](#)
[IUserUnit Members](#)

Availability

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SeparatorCharacter Property (IUserUnit)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserUnit Interface](#) : SeparatorCharacter Property (IUserUnit)

Gets the decimal separator character.

.NET Syntax

Visual Basic (Declaration)

Property SeparatorCharacter As System.String

Visual Basic (Usage)



```
Dim instance As IUserUnit
Dim value As System.String

instance.SeparatorCharacter = value
value = instance.SeparatorCharacter
```

C#

```
System.string SeparatorCharacter {get; set;}
```

C++/CLI

```
property System.String^ SeparatorCharacter {
    System.String^ get();
    void set ( & System.String^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Character passed as a string to represent the decimal separator

Visual Basic for Applications (VBA) Syntax

See [UserUnit.SeparatorCharacter](#).

Example

See the [IUserUnit](#) examples.

See Also

Availability

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SignificantDigits Property (IUserUnit)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserUnit Interface](#) : SignificantDigits Property (IUserUnit)

Gets the number of significant digits.

.NET Syntax

Visual Basic (Declaration)

Property SignificantDigits As System.Integer

Visual Basic (Usage)



```
Dim instance As IUserUnit
Dim value As System.Integer

instance.SignificantDigits = value
value = instance.SignificantDigits
```

C#

```
System.int SignificantDigits {get; set;}
```

C++/CLI

```
property System.int SignificantDigits {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Number of significant digits

Visual Basic for Applications (VBA) Syntax

See [UserUnit::SignificantDigits](#).

Example

See the [IUserUnit](#) examples.

See Also

[IUserUnit Interface](#)
[IUserUnit Members](#)

Availability

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SpecificUnitType Property (IUserUnit)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserUnit Interface](#) : SpecificUnitType Property (IUserUnit)

Gets the specific unit type for this user unit.

.NET Syntax

Visual Basic (Declaration)

```
Property SpecificUnitType As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IUserUnit
Dim value As System.Integer

instance.SpecificUnitType = value

value = instance.SpecificUnitType
```

C#

```
System.int SpecificUnitType {get; set;}
```

C++/CLI

```
property System.int SpecificUnitType {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

If [IUserUnit::UnitType](#) is [swUserUnitsType_e...](#)

Then this property is as defined in...

[swLengthUnit](#)

[swLengthUnit_e](#)

[swAngleUnit](#)

[swAngleUnit_e](#)

Visual Basic for Applications (VBA) Syntax

See [UserUnit::SpecificUnitType](#).

Example

See the [IUserUnit](#) examples.

See Also

[IUserUnit Interface](#)
[IUserUnit Members](#)

Availability

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UnitType Property (IUserUnit)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IUserUnit Interface](#) : UnitType Property (IUserUnit)

Gets the user unit type.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property UnitType As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IUserUnit  
Dim value As System.Integer  
  
value = instance.UnitType
```

C#

```
System.int UnitType {get;}
```

C++/CLI

```
property System.int UnitType {  
    System.int get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

User unit type as defined in [swUserUnitsType_e](#)

Visual Basic for Applications (VBA) Syntax

See [UserUnit::UnitType](#).

Example

See the [IUserUnit](#) examples.

See Also

[IUserUnit Interface](#)
[IUserUnit Members](#)

Availability

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